

Multiple Choice

1. **Answer:** A

Options: A) Human sperm cannot develop at the high temperature found within the body cavity.; B) The testes require the lower pressure environment of the scrotum for sperm maturation; C) Testes must be in the scrotum to properly attach to the vas deferens; D) The scrotum provides the necessary nutrients for sperm development; E) Gravity is required for the testes to function properly

Note: Correct option: A - Human sperm cannot develop at the high temperature found within the body cavity.

2. **Answer:** A

Options: A) Carbon dioxide bubbles forming in the blood causing acidosis; B) Rapid increase in barometric pressure causing the bends; C) Formation of helium bubbles in the bloodstream due to rapid ascent; D) Increased salinity in the water leading to dehydration and joint pain; E) Oxygen bubbles forming in the blood leading to hypoxia

Note: Correct option: A - Carbon dioxide bubbles forming in the blood causing acidosis

3. **Answer:** A

Options: A) Protects the cell from foreign invaders; B) Acts as a storage for genetic material; C) Regulates the pH level within the cell; D) Transports oxygen throughout the cell; E) Removes nitrogenous wastes

Note: Correct option: A - Protects the cell from foreign invaders

4. **Answer:** A

Options: A) The human genome has fully adapted to current environmental pressures; B) Modern medicine has eliminated the effects of genetic drift; C) The rate of genetic drift is directly proportional to technological advancements; D) Humans have reached a genetic equilibrium where drift is no longer possible; E) Humans have developed immunity to genetic drift

Note: Correct option: A - The human genome has fully adapted to current environmental pressures

5. **Answer:** B

Options: A) The absence of a Y chromosome determines maleness in mammals; B) The presence of a Y chromosome usually determines maleness in mammals; C) The ratio of two X chromosomes to one Y chromosome determines maleness; D) The presence of an X chromosome determines maleness in mammals; E) The presence of an X chromosome in the absence of a Y chromosome determines maleness

Note: Correct option: B - The presence of a Y chromosome usually determines maleness in mammals

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'The resulting colony is minute.'
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Chloroplast'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** (a) 0.0049, (b) 0.1302, (c) 0.8649. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** Transgressive segregation is when there is no variability in the F₂ generation.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key